

Module-3 (Assignment Day-3)

1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.



Steps:

- 1) Connect your computer to your home Wi-Fi.
- 2) Press Windows + R, type cmd, and press Enter.
- 3) Type: **ipconfig**
- 4) Find Default Gateway under your Wi-Fi adapter. It may be 192.168.0.1 or 192.168.1.1.
- 5) Open Chrome or Edge and enter that address in the address bar.
- 6) The router's admin login page should appear.
- 7) Take a screenshot using Windows + Shift + S.
- 8) Do not share your router password or any other private login details.

2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.



- **IP Address:** 192.168.31.102
- **Subnet mask:** 255.255.255.0
- **Default Gateway:** fe80::caa3:e8ff:fede:6df8%2
192.168.31.1

3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

➤

1. Current IP (DHCP)

```
C:\Windows\system32\cmd.exe
C:\Users\User>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : 
    IPv4 Address. . . . . : 192.168.1.101
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1
```

IP obtained automatically from router (DHCP)

2. Set Static IP

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports the capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 1 . 50
Subnet mask: 255 . 255 . 255 . 0
Default gateway: 192 . 168 . 1 . 1

Preferred DNS server: 8 . 8 . 8 . 8
Alternate DNS server: 8 . 8 . 4 . 4

☐ Validate settings upon exit

Advanced...

OK Cancel

Example Static IP (same range as router 192.168.1.1)

- IP: 192.168.1.50
- Subnet: 255.255.255.0
- Gateway: 192.168.1.1
- DNS: 8.8.8.8

3. Verify Static IP

```
C:\Windows\system32\cmd.exe
C:\Users\User>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : 
    IPv4 Address. . . . . : 192.168.1.50
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1
```

Static IP is now set.

4. Test Internet Access

```
C:\Windows\system32\cmd.exe
C:\Users\User>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=21ms TTL=115
Reply from 8.8.8.8: bytes=32 time=19ms TTL=115
Reply from 8.8.8.8: bytes=32 time=18ms TTL=115
Reply from 8.8.8.8: bytes=32 time=20ms TTL=115

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 18ms, Maximum = 21ms, Average = 19ms
```

Internet is working with Static IP

5. Revert to Automatic (DHCP)

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports the capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☒ Obtain an IP address automatically

☐ Use the following IP address:

IP address:
Subnet mask:
Default gateway:

Preferred DNS server:
Alternate DNS server:

☐ Validate settings upon exit

Advanced...

OK Cancel

After testing, select DHCP again to avoid IP conflicts.

6. Verify DHCP Again

```
C:\Windows\system32\cmd.exe
C:\Users\User>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : 
    IPv4 Address. . . . . : 192.168.1.101
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1
```

Back to DHCP (automatic IP).

IMPORTANT:
Always revert to automatic (DHCP) after testing to avoid IP conflicts on your network.



4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.



TL-WR840N x +

← → ↻ ⚠ Not secure | 192.168.1.1

tp-link TP-Link Wireless N Router WR840N
Model No. TL-WR840N

Navigation Menu:

- Status
- Quick Setup
- Operation Mode
- Network
- Wireless
- DHCP**
 - DHCP Settings
 - DHCP Clients List
 - Address Reservation
- Forwarding
- Security
- Parental Controls
- Access Control
- Advanced Routing
- Bandwidth Control
- IP & MAC Binding
- Dynamic DNS
- IPv6
- System Tools
- Logout

DHCP Settings

DHCP Server: ☐ Disable ☒ Enable

Start IP Address: 192.168.1.100

End IP Address: 192.168.1.199

Address Lease Time: 120 minutes (1-2880 minutes, the default value is 120)

Default Gateway: 192.168.1.1 (optional)

Default Domain: (optional)

Primary DNS: 8.8.8.8 (optional)

Secondary DNS: 8.8.4.4 (optional)

DHCP IP Address Range (Current Range)
192.168.1.100 – 192.168.1.199

Save

DHCP Settings Help

The Device is set up by default as a DHCP (Dynamic Host Configuration Protocol) server, which provides the TCP/IP configuration for all the PCs that are connected to the Device in the LAN.

- DHCP Server** - Enable or Disable the server. If you disable the Server, you must have another DHCP server within your network or else you must configure the computer manually.
- Start IP Address** - This field specifies the first address in the IP Address pool. 192.168.1.100 is the default start IP address.
- End IP Address** - This field specifies the last address in the IP Address pool. 192.168.1.199 is the default end IP address.
- Address Lease Time** - The Address Lease Time is the length of time a network user will be allowed to keep connecting to the Device with the current DHCP Address. Enter the amount of time, in minutes, that the DHCP address will be "leased". The time range is 1 - 2880 minutes. The default value is 120 minutes.

CURRENT DHCP RANGE (in my router)

192.168.1.100 – 192.168.1.199

- Start IP Address: 192.168.1.100
- End IP Address: 192.168.1.199
- Default Gateway: 192.168.1.1
- Subnet Mask: 255.255.255.0

What happens if two devices get the same IP?

- Both devices will try to use the same IP address.
- This causes an **IP address conflict**.
- One or both devices may lose internet connection or not work properly.
- You may face issues like intermittent disconnections, slow network, or websites not opening.

Tip: It is important that each device has a unique IP address in the network.